

Week 2

2.1 | The importance of data in AI applications



⌚ 30 minutes

The significance of data in AI applications is paramount, as it serves as the bedrock upon which AI systems learn, adapt, and make decisions. In essence, the effectiveness and accuracy of AI largely hinge on the quality, diversity, and volume of the data used for training machine learning models. This data-driven approach enables AI to identify patterns, make predictions, and facilitate decision-making processes in various fields, ranging from healthcare and finance to marketing and manufacturing. However, the quality of data also brings to the forefront issues of bias and ethical concerns. Poor or biased data can lead to inaccurate AI predictions or discriminatory outcomes, underlining the necessity for high-quality, diverse data sets in developing fair and effective AI systems.

Moreover, the role of data in AI extends beyond just model training to real-time decision-making and enhancing user experiences. AI systems, especially in applications like autonomous vehicles, rely on real-time data for immediate and safe operations. In consumer-focused services, AI uses data to personalise user experiences, as seen in recommendation systems on e-commerce platforms. Simultaneously, the emergence of big data has exponentially increased the potential of AI, unlocked new insights and enabling AI to drive innovation and operational efficiency in businesses. However, this heavy reliance on data also raises critical issues around data privacy and security, necessitating responsible and secure handling of sensitive information. As AI continues to evolve, managing data effectively and ethically remains crucial for leveraging the full potential of AI technologies.

The challenge

You will gain a foundational understanding of the data commonly used in AI, which is crucial for developing successful AI applications and recognising the ethical considerations involved in their implementation. Understanding data is not only just about technical issues (e.g. structured vs unstructured, quantitative vs qualitative) but also about other governance issues (e.g., data governance, data quality).

The output



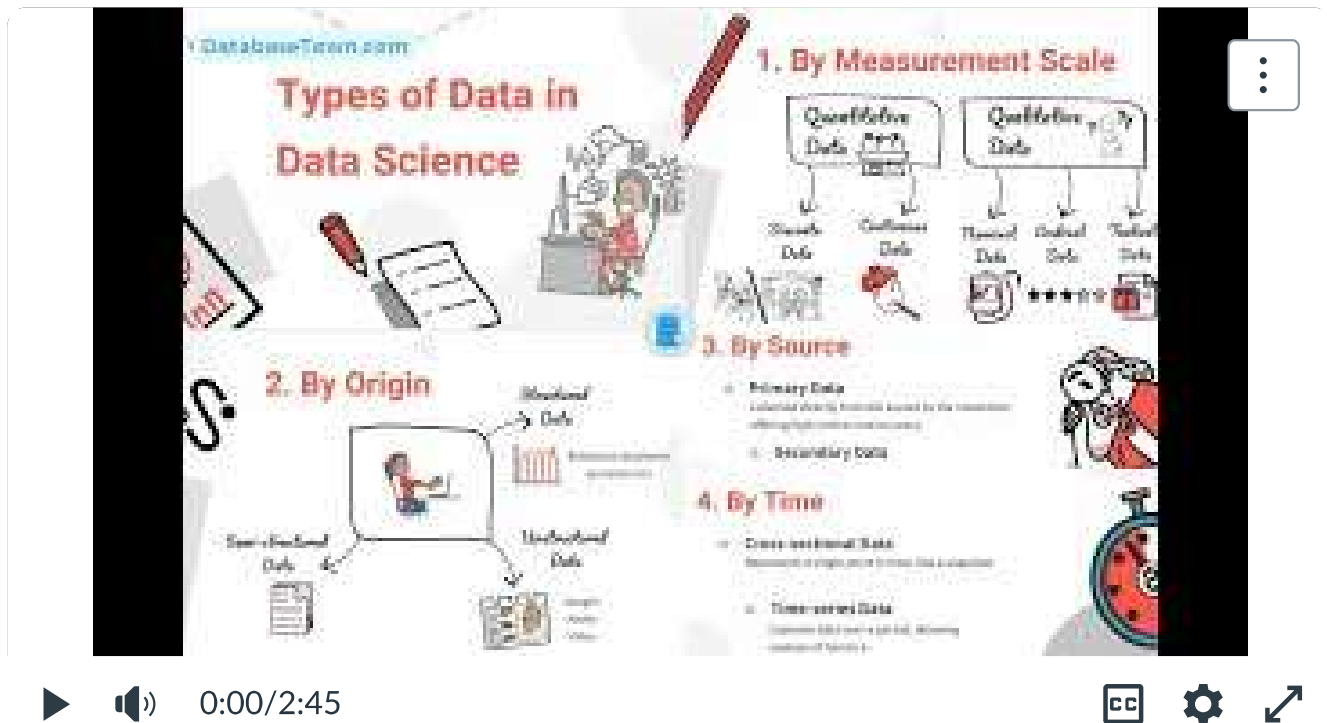
A completed quiz.

The task

In this task, you will gain a basic understanding of data and AI through a series of learning materials, then complete a quiz based on concepts explored in the task materials.

How can we categorise data?

Before starting your reading, please watch this short video to understand basic ways to categorise data (and standards used to refer to certain types of data):



How is data used in AI applications?

Read the following two articles to gain basic a understanding of the role of data in AI applications and practical concerns:

- [If Your Data Is Bad, Your Machine Learning Tools Are Useless](#) ➞ (Redman, 2018)
- [The Role of Data in AI: Report for the Data Governance](#) ➞ (GPAI, 2020)

Recap: Compare AI, machine learning, and analytics

Review the following table for a quick comparison of AI, machine learning, and analytics as three approaches to extract values from data:

Aspect	Artificial Intelligence (AI)	Machine Learning (ML)	Analytics
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Definition	A broad concept that aims to create machines capable of human-like intelligence and reasoning across various tasks.	A subset of AI focused on developing algorithms that learn from data to make predictions or decisions.	The process of examining data to draw conclusions or make informed decisions.
Goal	To replicate human cognitive functions and adapt to different domains, exhibiting intelligence.	To improve performance on specific tasks through data-driven learning.	To analyse data for insights, trends, and informed decision-making.
Learning Approach	May or may not involve learning from data; can include rule-based and knowledge-based systems.	Primarily data-driven, relies on training algorithms with data to learn patterns.	Typically, does not involve learning but focuses on data exploration and reporting.
Types	Encompasses a wide range of approaches, including rule-based systems, expert systems, symbolic AI, and machine learning.	Categorised into supervised learning, unsupervised learning, reinforcement learning, and more. (We will explore more in the next challenge)	Includes descriptive analytics, diagnostic analytics, predictive analytics, and prescriptive analytics.

Examples

Virtual assistants (e.g., Siri, Alexa), self-driving cars, game-playing AI (e.g., AlphaGo), natural language understanding systems.

Spam email detection, recommendation systems (e.g., Netflix, Amazon), image and speech recognition, autonomous robotics.

Generating reports, dashboards, data visualisation, market research, business intelligence tools.

Instructions

Review the task materials, then complete the following quiz:

Your progress was restored [Start over](#)



What is the primary reason for not deleting training data after an AI model is developed, as mentioned in the text?

- ☐ It prevents data misuse
- ☐ Legal obligations require data retention
- ☐ Re-training might be needed
- ☐ It reduces the cost of data storage

✓ Check



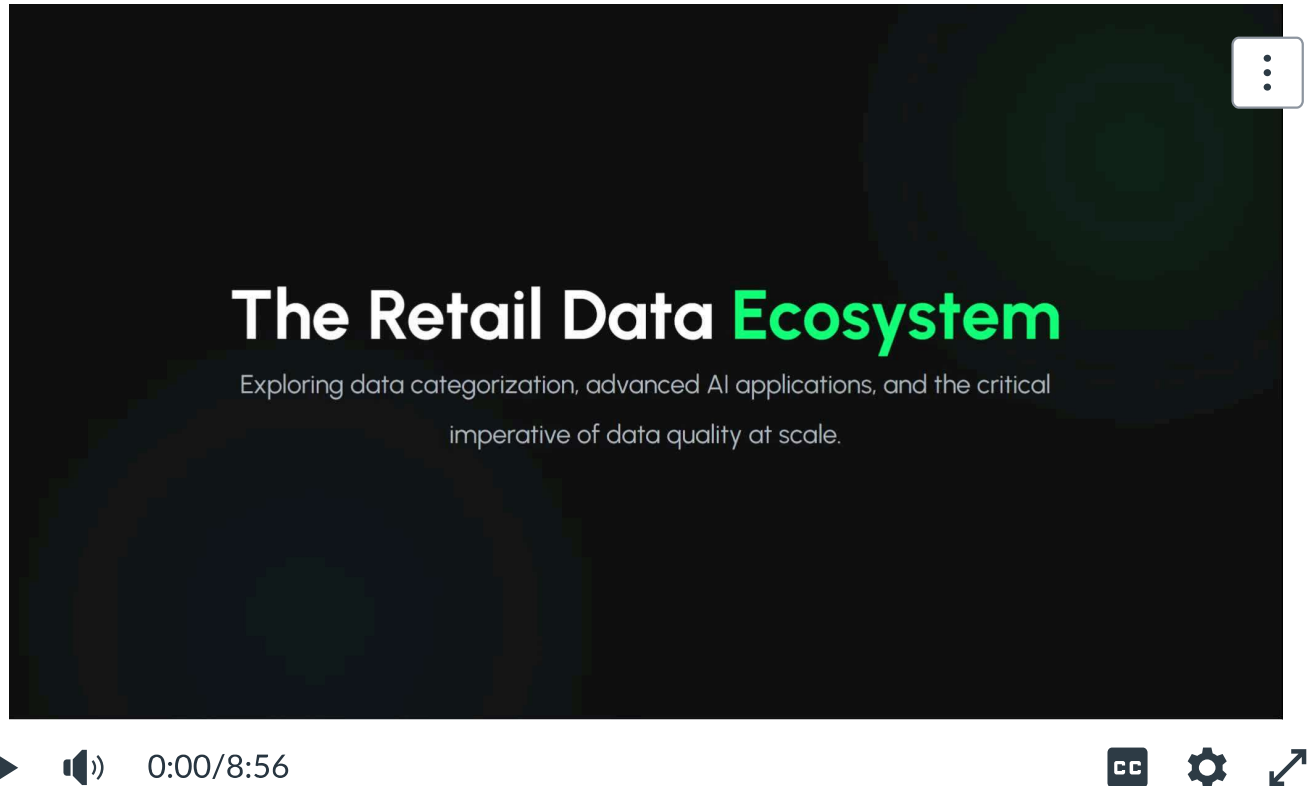
Reuse

I am confused

Come prepared to discuss concepts from the quiz in class.



Watch: The Retail Data Ecosystem & AI Strategy



Supporting resources

The following reading materials will be very helpful to help you gain basic understanding of how data is created, collected, and used in AI applications:

- [Liang, W., Tadesse, G.A., Ho, D. et al. Advances, challenges and opportunities in creating data for trustworthy AI](#) . Nat Mach Intell 4, 669–677 (2022).